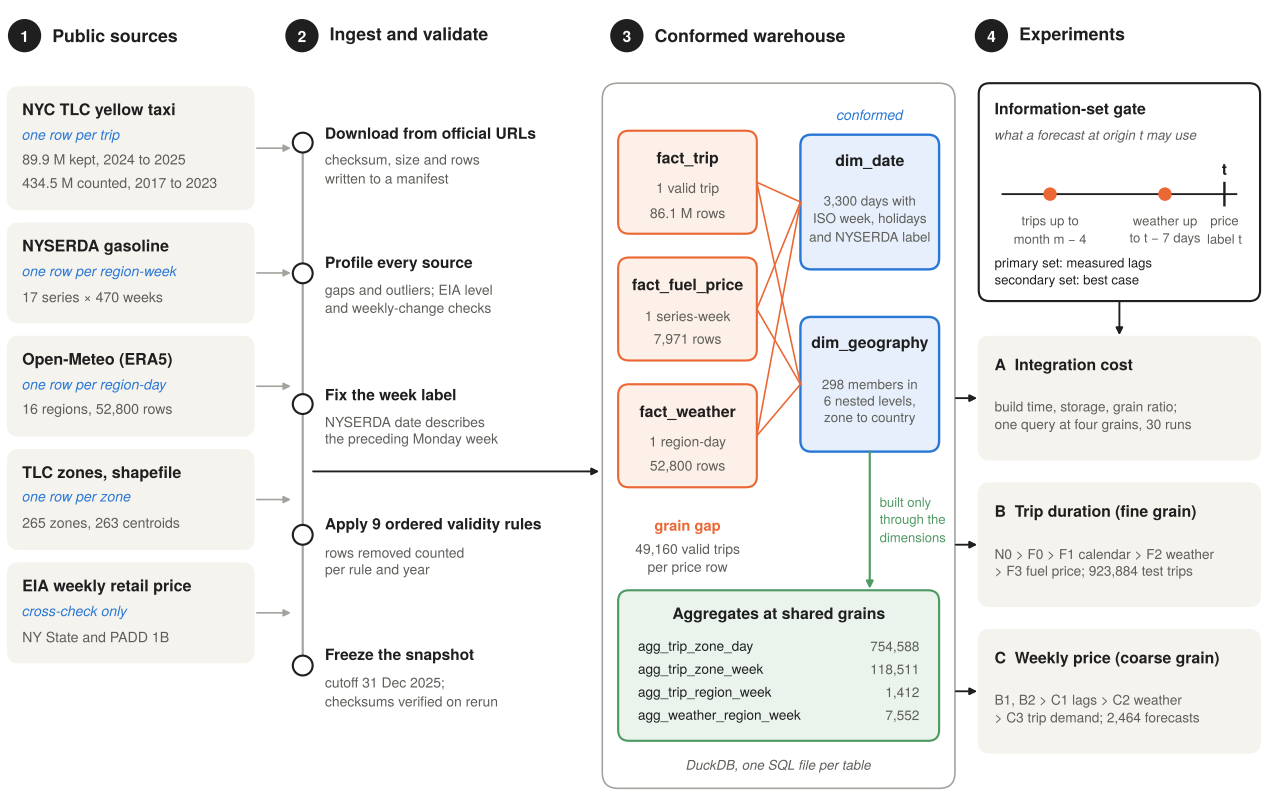
1 Public sources
2 Ingest and validate
3 Conformed warehouse
4 Experiments
NYC TLC yellow taxi
one row per trip
89.9 M kept, 2024 to 2025
434.5 M counted, 2017 to 2023
NYSERDA gasoline
one row per region-week
17 series × 470 weeks
Open-Meteo (ERA5)
one row per region-day
16 regions, 52,800 rows
TLC zones, shapefile
one row per zone
265 zones, 263 centroids
EIA weekly retail price
cross-check only
NY State and PADD 1B
Download from official URLs
checksum, size and rows written to a manifest
Profile every source
gaps and outliers; EIA level and weekly-change checks
Fix the week label
NYSERDA date describes the preceding Monday week
Apply 9 ordered validity rules
rows removed counted per rule and year
Freeze the snapshot
cutoff 31 Dec 2025; checksums verified on rerun
conformed
fact_trip
1 valid trip
86.1 M rows
fact_fuel_price
1 series-week
7,971 rows
fact_weather
1 region-day
52,800 rows
dim_date
3,300 days with ISO week, holidays and NYSERDA label
dim_geography
298 members in 6 nested levels, zone to country
grain gap
49,160 valid trips per price row
built only through the dimensions
Aggregates at shared grains
agg_trip_zone_day 754,588
agg_trip_zone_week 118,511
agg_trip_region_week 1,412
agg_weather_region_week 7,552
DuckDB, one SQL file per table
Information-set gate
what a forecast at origin t may use
t
trips up to month m − 4
weather up to t − 7 days
price label t
primary set: measured lags
secondary set: best case
A Integration cost
build time, storage, grain ratio; one query at four grains, 30 runs
B Trip duration (fine grain)
N0 > F0 > F1 calendar > F2 weather > F3 fuel price; 923,884 test trips
C Weekly price (coarse grain)
B1, B2 > C1 lags > C2 weather > C3 trip demand; 2,464 forecasts